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Display device

Abstract

A display device that can improve transmittance of a sensor area that overlaps a display area includes a substrate that includes a display area in which a plurality of pixels are disposed, a sensor area in the display area, the sensor area overlapping a sensor, and a wiring connection area between the display area and the sensor area; a first wiring and a second wiring disposed in the display area and that extend in a first direction and are connected to the plurality of pixels; and a third wiring disposed in the sensor area and that extends in the first direction, wherein the third wiring is connected to the second wiring and overlaps the first wiring in a plan view. The third wiring is spaced apart from the first wiring with a first insulating layer interposed therebetween.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 16/139,739, filed on Sep. 24, 2018 in the U.S. Patent and Trademark Officer, which claims priority under 35 U.S.C. § 119 from, and the benefit of, Korean Patent Application No. 10-2017-0132099, filed on Oct. 12, 2017 in the Korean Intellectual Property Office (KIPO), the contents of both of which are herein incorporated by reference in their entireties.

TECHNICAL FIELD
(1) Exemplary embodiments of the present invention are directed to a display device that can improve transmittance of a sensor area in a display area.
DISCUSSION OF RELATED ART
(2) Flat panel display (FPD) devices have reduced weight and volume, as compared to a cathode ray tube (CRT). Such FPD devices include liquid crystal display (LCD) devices, field emission display (FED) devices, plasma display panel (PDP), devices or organic light emitting diode (OLED) display devices, for example.
(3) Among the FPD devices, OLED display devices display images use OLEDs to generate light by recombination of electrons and holes.
(4) In addition, as a display area occupies most of a front surface in a mobile terminal, other components, such as a camera, a proximity sensor, a fingerprint sensor, an illuminance sensor, a near-infrared sensor, etc., may overlap the display area.
SUMMARY
(5) Exemplary embodiments of the present invention are directed to a display device that can improve transmittance of a sensor area that overlaps a display area.
(6) According to an exemplary embodiment, a display device includes: a substrate that includes a display area in which a plurality of pixels are disposed, a sensor area in the display area, the sensor area overlapping a sensor, and a wiring connection area between the display area and the sensor area; a first wiring and a second wiring disposed in the display area that extend in a first direction and are connected to the plurality of pixels; and a third wiring disposed in the sensor area and that extends in the first direction, wherein the third wiring is connected to the second wiring and overlaps the first wiring in a plan view. The third wiring is spaced apart from the first wiring with a first insulating layer interposed therebetween.
(7) A contact hole may penetrate the first insulating layer in the wiring connection area through which second wiring is connected to the third wiring.
(8) The third wiring may further include a first bent portion in the wiring connection area that extends in a second direction that crosses the first direction wherein the first bent portion connects

to the second wiring through the contact hole.
(9) The first wiring may extend in the first direction into the sensor area.
(10) The display device may further include a plurality of dummy pixels in the wiring connection area adjacent to the display area.
(11) The first wiring and the second wiring may be disposed on a substantially same layer.
(12) The second wiring and the third wiring may be disposed on a substantially same layer.
(13) The third wiring may further include a second bent portion in the wiring connection area that extends in the second direction.
(14) The third wiring may overlap the first wiring in the wiring connection area and the sensor area in a plan view.
(15) The display device may further include a fifth wiring connected to the pixel and that extends in a second direction that crosses the first, second and third wirings. The fifth wiring may be spaced apart from the first, second and third wirings by a second insulating layer.
(16) The fifth wiring may be disposed above the third wiring.
(17) The fifth wiring may be disposed below the third wiring.
(18) The pixel may include a pixel electrode on the substrate, a light emitting layer on the pixel electrode and a common electrode on the light emitting layer. One of the first wiring, the third wiring or the fifth wiring may be disposed in in the sensor area in substantially the same layer as the pixel electrode. At least one of the pixel electrode, the light emitting layer or the common electrode is not disposed in the sensor area
(19) One of the first wiring, the third wiring or the fifth wiring may be a transparent electrode or a nanowire in a form of a mesh in the sensor area.
(20) The display device may further include at least one floating wiring that overlaps the fifth wiring in a plan view and that is connected to the fifth wiring through a contact hole.
(21) According to an exemplary embodiment, a display device includes: a substrate that includes a display area in which a plurality of pixels are disposed and a sensor area in the display area, the sensor area overlapping a sensor; and a gate wiring, a data wiring and a power wiring connected to the pixel. The power wiring has a mesh shape in which a plurality of wirings cross each other in a plane and are not disposed in the sensor area.
(22) The power wiring may include a first power wiring that extends in a first direction and a second power wiring that extends in a second direction that crosses the first direction.
(23) The first power wiring and the second power wiring may be disposed on a substantially same layer.
(24) The first power wiring and the second power wiring may be spaced apart from each other by a first insulating layer and be connected to each other through a contact hole formed in the first insulating layer.
(25) The first power wiring and the gate wiring may be disposed in substantially a same layer and the second power wiring and the data wiring may be disposed in substantially a same layer.
(26) At least one of the gate wiring or the data wiring may cross the sensor area.
(27) The gate wiring may include a first gate wiring and a second gate wiring. The first gate wiring and the second gate wiring may be spaced apart from each other in the display area and overlap each other in a plan view in the sensor area with a second insulating layer therebetween.
(28) Each of the plurality of pixels may include a pixel electrode connected to the data wiring, a light emitting layer on the pixel electrode and a common electrode on the light emitting layer. One of the data wiring, the first gate wiring or the second gate wiring may be disposed in the sensor area in substantially a same layer as the pixel electrode.
(29) One of the data wiring, the first gate wiring or the second gate wiring may be a transparent electrode or a nanowire in a form of a mesh in the sensor area.
(30) At least one of the gate wiring or the data wiring is not disposed inh the sensor area.
(31) At least one of the gate wiring or the data wiring may include: a first wiring and a second

wiring disposed in the display area and that extend in a first direction across the sensor area and are separated from each other by the sensor area; and a third wiring disposed outside the sensor area and that connects the first wiring and the second wiring.
(32) The third wiring may be spaced apart from the first wiring and the second wiring with a third insulating layer interposed therebetween and the third wiring is connected to the first wiring and the second wiring through a contact hole formed in the third insulating layer.
(33) The display device may further include a sealing member on the substrate, the sealing member including one or more inorganic layers and one or more organic layers which are alternately disposed. The organic layer may be disposed outside the sensor area.
(34) The display device may further include a touch electrode on the substrate. The touch electrode may be disposed outside the sensor area.
(35) According to an exemplary embodiment, a display device includes: a substrate that includes a display area in which a plurality of pixels are disposed and a sensor area in the display area, the sensor area overlapping a sensor; a wiring connected to the pixel, the wiring that crosses the sensor area; and at least one of an insulating layer above the wiring or an insulating layer below the wiring. The insulating layer is not disposed in an area that does not overlap the wiring in the sensor area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a block diagram of a display device according to an exemplary embodiment.
(2) FIG. **2** is an equivalent circuit diagram of one pixel illustrated in FIG. **1**.
(3) FIG. **3** is a detailed plan view of a display device that includes a pixel of FIG. **1** and lines connected thereto.
(4) FIGS. **4**A, **4**B, **4**C, **4**D, **4**E and **4**F separately illustrate a part of the components of FIG. **3**.
(5) FIG. **5** is a cross-sectional view taken along line I-I′ of FIG. **3**.
(6) FIG. **6** is a plan view of a display device according to an exemplary embodiment.
(7) FIG. **7** is an enlarged view of an area where a wiring connection area meets a sensor area in an X-axis direction, according to an exemplary embodiment.
(8) FIGS. **8**, **9** and **10** are cross-sectional views taken along line I-I′ of FIG. **7** according to an exemplary embodiment.
(9) FIG. **11** is a cross-sectional view taken along line of FIG. **7**.
(10) FIG. **12** is an enlarged view of an area where a wiring connection area meets a sensor area in the X-axis direction, according to an exemplary embodiment;
(11) FIG. **13** is a cross-sectional view taken along line I-I′ of FIG. **12**, according to an exemplary embodiment.
(12) FIG. **14** is a wiring diagram of a high potential line around a sensor area, according to an exemplary embodiment.
(13) FIG. **15** is an enlarged view of an area where a sensor area meets a display area in a Y-axis direction, according to an exemplary embodiment.
(14) FIG. **16** is a cross-sectional view taken along line I-I′ of FIG. **15**.
(15) FIG. **17** is an enlarged view of an area where a sensor area meets a display area in a Y-axis direction according to an exemplary embodiment.
(16) FIG. **18** is a cross-sectional view taken along line I-I′ of FIG. **17**.
(17) FIGS. **19** and **20** are wiring diagrams of display devices according to exemplary embodiments.
(18) FIGS. **21**A, **21**B, **21**C and **21**D are cross-sectional views of a touch panel and a sealing member of a display device according to an exemplary embodiment.
DETAILED DESCRIPTION

(19) Exemplary embodiments will now be described more fully hereinafter with reference to the accompanying drawings. Although embodiments may be modified in various manners and have several exemplary embodiments, exemplary embodiments are illustrated in the accompanying drawings and will be mainly described in the specification. However, the scope of the invention is not limited to the exemplary embodiments and should be construed as including all the changes, equivalents and substitutions included in the spirit and scope of the invention.
(20) In the drawings, thicknesses of a plurality of layers and areas may be illustrated in an enlarged manner for clarity and ease of description thereof. When a layer, area, or plate is referred to as being "on" another layer, area, or plate, it may be directly on the other layer, area, or plate, or intervening layers, areas, or plates may be present therebetween. Further when a layer, area, or plate is referred to as being "below" another layer, area, or plate, it may be directly below the other layer, area, or plate, or intervening layers, areas, or plates may be present therebetween.
(21) Throughout the specification, when an element is referred to as being "connected" to another element, the element may be "directly connected" to the other element, or "electrically connected" to the other element with one or more intervening elements interposed therebetween.
(22) It will be understood that the words "about" or "approximately" as used herein is inclusive of the stated value and means within an acceptable range of deviation for the particular value as determined by one of ordinary skill in the art, considering the measurement in question and the error associated with measurement of the particular quantity, i.e., the limitations of the measurement system.
(23) Hereinafter, a display device according to an exemplary embodiment will be described in detail with reference to FIGS. **1** to **21**D.
(24) FIG. **1** is a block diagram of a display device according to an exemplary embodiment.
(25) A display device **100** according to an exemplary embodiment includes a display panel **101**, a scan driver **102**, a light emission control driver **103**, a data driver **104** and a power supply portion **105**, as illustrated in FIG. **1**.
(26) According to an embodiment, the display panel **101** includes "i+2" number of scan lines SL**0** to SLi+1, "k" number of light emission control lines EL**1** to ELk, "j" number of data lines DL**1** to DLj and "k×j" number of pixels PX, where each of i, j and k is a natural number greater than 1.
(27) According to an embodiment, the scan lines SL**0** to SLi+1 are spaced apart from each other in a Y-axis direction and each of the scan lines SL**0** to SLi+1 extends in an X-axis direction. The light emission control lines EL**1** to ELk are spaced apart from each other in the Y-axis direction and each of the light emission control lines EL**1** to ELk extends in the X-axis direction. The data lines DL**1** to DLj are spaced apart from each other in the X-axis direction and each of the data lines DL**1** to DLj extends in the Y-axis direction.
(28) According to an embodiment, the scan driver **102** generates scan signals based on a scan control signal received from a timing controller and sequentially transmits the scan signals to the plurality of scan lines SL**0** to SLi+1. During one frame period, the scan driver **102** transmits the first to i-th scan signals sequentially starting from the first scan signal.
(29) According to an embodiment, the light emission control driver **103** generates light emission control signals based on a control signal received from a timing controller and sequentially transmits the light emission control signals to the plurality of light emission control lines EL**1** to ELk. In an exemplary embodiment, the light emission control driver **103** is embedded in the scan driver **102**. For example, the scan driver **102** further performs the function of the light emission control driver **103**. In such an exemplary embodiment, the scan lines SL**0** to SLi+1 and the light emission control lines EL**1** to ELk are driven together by the scan driver **102**. Unless otherwise specifically defined, the scan lines and the scan driver may be understood to include the light emission control lines and the light emission control driver, respectively.
(30) According to an embodiment, the data driver **104** transmits first to i-th data voltages to the first to j-th data lines DL**1** to DLj, respectively. For example, the data driver **104** receives image data

signals and data control signals from a timing controller. In addition, the data driver **104** samples the image data signals based on the data control signal, latches the sampled image data signals sequentially to correspond to one horizontal line in each horizontal period and transmits the latched image data signals to the data lines DL**1** to DLj substantially simultaneously.
(31) According to an embodiment, the pixels PX are arranged at the display panel **101** in a matrix form. The pixels PX are disposed in a display area **310**, shown in FIG. **6**, of the display panel **101**. The pixels PX emit light of different colors. For example, of pixels PX illustrated in
(32) FIG. **1**, a pixel indicated by a symbol "R" emits red light, a pixel indicated by a symbol "G" emits green light and a pixel indicated by a symbol "B" emits blue light. In an exemplary embodiment, the display panel **101** may further include at least one white pixel which emits white light.
(33) According to an embodiment, each pixel PX receives a high potential driving voltage ELVDD, a low potential driving voltage ELVSS and an initialization voltage Vinit from the power supply portion **105**.
(34) FIG. **2** is an equivalent circuit diagram of one pixel illustrated in FIG. **1**, according to an exemplary embodiment.
(35) As illustrated in FIG. **2**, according to an embodiment, one pixel PX includes a first thin film transistor T**1**, a second thin film transistor T**2**, a third thin film transistor T**3**, a fourth thin film transistor T**4**, a fifth thin film transistor T**5**, a sixth thin film transistor T**6**, a seventh thin film transistor T**7**, a storage capacitor Cst and an organic light emitting diode ("OLED").
(36) According to an embodiment, each of the first, second, third, fourth, fifth, sixth and seventh thin film transistors T**1**, T**2**, T**3**, T**4**, T**5**, T**6** and T**7** is a P-type thin film transistor, as illustrated in FIG. **2**. However, exemplary embodiments are not limited thereto and in an exemplary embodiment, each of the first, second, third, fourth, fifth, sixth and seventh thin film transistors T**1**, T**2**, T**3**, T**4**, T**5**, T**6** and T**7** may be an N-type thin film transistor.
(37) FIG. **3** is a detailed plan view of a display device that includes a pixel of FIG. **1** and lines connected thereto, FIGS. **4**A, **4**B, **4**C, **4**D, **4**E and **4**F separately illustrate a part of the components of FIG. **3** and FIG. **5** are cross-sectional views taken along the line I-I′ of FIG. **3**.
(38) According to an embodiment, FIG. **4**A illustrates a semiconductor layer **130** of FIG. **3**, FIG. **4**B illustrates an (n−1)-th scan line SLn−1, an n-th scan line SLn, an (n+1)-th scan line SLn+1 and a light emission control line EL, FIG. **4**C illustrates an initialization line IL and a capacitor electrode **171** of FIG. **3**, FIG. **4**D illustrates the data line DL and the high potential line VDL of
(39) FIG. **3**, FIG. **4**E illustrates a pixel electrode PE of FIG. **3** and FIG. **4**F illustrates the semiconductor layer **130**, the (n−1)-th scan line SLn−1, the n-th scan line SLn, the (n+1)-th scan line SLn+1 and the light emission control line EL of FIG. **3**.
(40) As illustrated in FIGS. **3**, **4***b* and **5**, a display device according to an exemplary embodiment includes a substrate **110**, a buffer layer **120**, a semiconductor layer **130**, a gate insulating layer **140**, gate electrodes GE**1**, GE**2**, GE**3**, GE**4**, GE**5**, GE**6** and GE**7**, an (n−1)-th scan line SLn−1, an n-th scan line SLn, an (n+1)-th scan line SLn+1, a light emission control line EL, a first insulating interlayer **160**, an initialization line IL, a capacitor electrode **171**, a second insulating interlayer **180**, a first connection electrode **181**, a second connection electrode **182**, a third connection electrode **183**, a data line DL, a high potential line VDL, a planarization layer **220**, a pixel electrode PE, a light blocking layer **240**, a light emitting layer **250**, a common electrode **260** and a sealing member **270**.
(41) According to an embodiment, the substrate **110** illustrated in FIG. **5** is a transparent insulating substrate that includes glass or transparent plastic. For example, the substrate **110** may include one of kapton, polyethersulphone (PES), polycarbonate (PC), polyimide (PI), polyethyleneterephthalate (PET), polyethylene naphthalate (PEN), polyacrylate (PAR), fiber reinforced plastic (FRP), etc.
(42) According to an embodiment, as illustrated in FIG. **5**, the buffer layer **120** is disposed on the substrate **110**. The buffer layer **120** is disposed over an entire surface of the substrate **110**. The

buffer layer **120** can prevent permeation of undesirable elements and planarizes a surface therebelow and includes materials suitable for planarizing or preventing permeation. For example, the buffer layer **120** may include one of a silicon nitride (SiN.sub.x) layer, a silicon oxide (SiO.sub.2) layer or a silicon oxynitride (SiO.sub.xN.sub.y) layer. However, in other embodiments the buffer layer **120** may be omitted depending on the type of the substrate **110** and process conditions thereof.

(43) According to an embodiment, as illustrated in FIG. **5**, the semiconductor layer **130** is disposed on the buffer layer **120**.

(44) According to an embodiment, as illustrated in FIG. **4**A, the semiconductor layer **130** provides respective channel areas CH**1**, CH**2**, CH**3**, CH**4**, CH**5**, CH**6** and CH**7** of the first, second, third, fourth, fifth, sixth and seventh thin film transistors T**1**, T**2**, T**3**, T**4**, T**5**, T**6** and T**7**. In addition, the semiconductor layer **130** provides source electrodes SE**1**, SE**2**, SE**3**, SE**4**, SE**5**, SE**6** and SE**7** and drain electrodes DE**1**, DE**2**, DE**3**, DE**4**, DE**5**, DE**6** and DE**7** of respective first, second, third, fourth, fifth, sixth and seventh thin film transistors T**1**, T**2**, T**3**, T**4**, T**5**, T**6** and T**7**.

(45) According to an embodiment, the semiconductor layer **130** includes at least one of a polycrystalline silicon layer, an amorphous silicon layer or an oxide semiconductor such as indium gallium zinc oxide (IGZO) or indium zinc tin oxide (IZTO). For example, when the semiconductor layer **130** includes a polycrystalline silicon layer, the semiconductor layer **130** includes a channel area which is not doped with impurities and source and drain electrodes formed on opposite sides of the channel area and doped with impurity ions.

(46) According to an embodiment, as illustrated in FIG. **5**, the gate insulating layer **140** is disposed on the semiconductor layer **130** and the buffer layer **120**. The gate insulating layer **140** includes at least one of tetraethylorthosilicate (TEOS), silicon nitride (SiN.sub.x) or silicon oxide (SiO.sub.2). For example, the gate insulating layer **140** may have a double-layer structure where an about 40 nm thick SiN.sub.x layer is sequentially stacked on an about 80 nm thick TEOS layer.

(47) According to an embodiment, as illustrated in FIG. **5**, the first gate electrode GE**1** is disposed on the gate insulating layer **140**. The first gate electrode GE**1** is positioned between the gate insulating layer **140** and the first insulating interlayer **160**.

(48) According to an embodiment, the scan lines and the light emission control line are disposed on the gate insulating layer. For example, the (n−1)-th scan line SLn−1, the n-th scan line SLn, the (n+1)-th scan line SLn+1 and the light emission control line EL are positioned between the gate insulating layer **140** and the first insulating interlayer **160**.

(49) According to an embodiment, the scan line SL (e.g., at least one of the (n−1)-th scan line SLn−1, the n-th scan line SLn, the (n+1)-th scan line SLn+1) include aluminum (Al) or alloys thereof, silver (Ag) or alloys thereof, copper (Cu) or alloys thereof, or molybdenum (Mo) or alloys thereof. Alternatively, the scan line SL includes one of chromium (Cr) or tantalum (Ta). In an exemplary embodiment, the scan line SL has a multilayer structure that includes at least two conductive layers that have different physical properties.

(50) In addition, according to an embodiment, the light emission control line EL includes substantially a same material and have substantially a same structure, such as a multilayer structure, as those of the scan line SL. The light emission control line EL and the scan line SL substantially simultaneously formed in substantially a same process.

(51) According to an embodiment, as illustrated in FIG. **5**, the first insulating interlayer **160** is disposed on the first gate electrode GE**1** and the gate insulating layer **140**. The first insulating interlayer **160** has a thickness greater than a thickness of the gate insulating layer **140**. The first insulating interlayer **160** includes a material that is substantially the same as a material included in the gate insulating layer **140** described above.

(52) According to an embodiment, as illustrated in FIG. **5**, the capacitor electrode **171** is disposed on the first insulating interlayer **160**. The capacitor electrode **171** together with the first gate electrode GE**1** described above form a storage capacitor Cst. As illustrated in FIGS. **3** and **4**C, the

capacitor electrode **171** has a hole **30**.

(53) According to an embodiment, the initialization line IL, shown in FIGS. **3** and **4**C, is also disposed on the first insulating interlayer **160**. For example, the initialization line IL is located between the first insulating interlayer **160** and the second insulating interlayer **180**.

(54) According to an embodiment, as illustrated in FIG. **5**, the second insulating interlayer **180** is disposed on the capacitor electrode **171** and the first insulating interlayer **160**. The second insulating interlayer **180** has a thickness greater than a thickness of the gate insulating layer **140**. The second insulating interlayer **180** includes a material substantially the same as a material included in the gate insulating layer **140** described above.

(55) According to an embodiment, as illustrated in FIG. **5**, the first connection electrode **181**, the second connection electrode **182**, the high potential line VDL and the data line DL are disposed on the second insulating interlayer **180**.

(56) According to an embodiment, the data line DL includes a refractory metal, such as molybdenum, chromium, tantalum and titanium, or an alloy thereof. The data line DL has a multilayer structure that includes a refractory metal layer and a low-resistance conductive layer. Examples of a multilayer structure include a double-layer structure that includes a chromium or molybdenum (alloy) lower layer and an aluminum (alloy) upper layer; or a triple-layer structure that includes a molybdenum (alloy) lower layer, an aluminum (alloy) intermediate layer and a molybdenum (alloy) upper layer. In an exemplary embodiment, the data line DL can include or be formed of any suitable metal or conductor other than the aforementioned materials.

(57) According to an embodiment, as illustrated in FIG. **5**, the planarization layer **220** is disposed on the first connection electrode **181**, the second connection electrode **182**, the high potential line VDL and the data line DL.

(58) According to an embodiment, the planarization layer **220** can planarize the substrate **110** by eliminating height differences of the substrate **110** to increase luminous efficiency of the OLED to be formed thereon. The planarization layer **220** includes one or more of the following materials: a polyacrylate resin, an epoxy resin, a phenolic resin, a polyamide resin, a polyimide resin, an unsaturated polyester resin, a polyphenylen ether resin, a polyphenylene sulfide resin or benzocyclobutene (BCB).

(59) According to an embodiment, as illustrated in FIG. **5**, the pixel electrode PE is disposed on the planarization layer **220**. At least a portion of the pixel electrode PE is disposed in a light emission area **280**. That is, the pixel electrode PE is positioned to correspond to a light emission area **280** defined by the light blocking layer **240**, described below. The pixel electrode PE is connected to the first connection electrode **181** through a contact hole **19** that penetrates through the planarization layer **220**.

(60) According to an embodiment, as illustrated in FIG. **5**, the light blocking layer **240** is located on the pixel electrode PE and the planarization layer **220**. The light blocking layer **240** has an opening therethrough that corresponds to the light emission area **280**.

(61) According to an embodiment, the light blocking layer **240** includes a resin such as a polyacrylate resin or a polyimide resin.

(62) According to an embodiment, the light emitting layer **250** is disposed on the pixel electrode PE in the light emission area **280** and the common electrode **260** is disposed on the light blocking layer **240** and the light emitting layer **250**. The pixel electrode PE, the light emitting layer **250** and the common electrode **260** constitute a light emitting diode, such as the OLED of FIG. **2**. In such an exemplary embodiment, the pixel electrode PE corresponds to an anode electrode of the OLED and the common electrode **260** corresponds to a cathode electrode of the OLED.

(63) According to an embodiment, the light emitting layer **250** may include a low molecular weight organic material or a high molecular weight organic material. In addition, at least one of a hole injection layer HIL or a hole transport layer HTL can be further provided between the pixel electrode PE and the light emitting layer **250** and at least one of an electron transport layer ETL or

an electron injection layer EIL can be further provided between the light emitting layer **250** and the common electrode **260**.
(64) According to an embodiment, the pixel electrode PE and the common electrode **260** can be one of a transmissive electrode, a transflective electrode or a reflective electrode.
(65) According to an embodiment, a transparent conductive oxide ("TCO") is used to form a transmissive electrode. A TCO includes at least one of indium tin oxide (ITO), indium zinc oxide (IZO), antimony tin oxide (ATO), aluminum zinc oxide (AZO), zinc oxide (ZnO) or mixtures thereof
(66) According to an embodiment, a metal, such as magnesium (Mg), silver (Ag), gold (Au), calcium (Ca), lithium (Li), chromium (Cr), aluminum (Al) or copper (Cu), or an alloy thereof, is used to form a transflective electrode and a reflective electrode. In such an exemplary embodiment, whether an electrode is a transflective type or a reflective type depends on the thickness of the electrode. Typically, a transflective electrode has a thickness of about 200 nm or less and a reflective electrode has a thickness of about 300 nm or more. As the thickness of a transflective electrode decreases, light transmittance increases. On the contrary, as the thickness of the transflective electrode increases, light transmittance decreases.
(67) In addition, according to an embodiment, a transflective electrode and a reflective electrode have a multilayer structure that includes a metal layer and a TCO layer stacked thereon.
(68) According to an embodiment, the sealing member **270** is disposed on the common electrode **260**. The sealing member **270** includes a transparent insulating material such as glass, transparent plastic, etc. In addition, the sealing member **270** has a thin film encapsulation structure in which one or more inorganic layers and one or more organic layers are alternately laminated along a Z-axis direction.
(69) In an exemplary embodiment, as illustrated in FIG. **5**, a plurality of patterns are disposed vertically on different layers between the substrate **110** and the pixel electrodes PE. For example, the plurality of patterns are disposed on different layers along the Z-axis direction between the substrate **110** and the pixel electrodes PE, which will be described below in detail.
(70) According to an embodiment, the semiconductor layer **130** of FIG. **4**A is a first pattern **130**, each of the scan lines SLn−1, SLn and SLn+1, each of the gate electrodes GE**1**, GE**2**, GE**3**, GE**4**, GE**5**, GE**6** and GE**7** and the light emission control line EL of FIG. **4**B form a second pattern **150**, the initialization line IL and the capacitor electrode **171** of FIG. **4**C form a third pattern **170**, each of the connection electrodes **181**, **182** and **183**, the high potential line VDL and the data line DL of FIG. **4**D form a fourth pattern **190**, and the pixel electrode PE of FIG. **4**E forms a fifth pattern **230**. The gate insulating layer **140**, the first insulating interlayer **160**, the second insulating interlayer **180** and the planarization layer **220** are disposed between each of the corresponding patterns.
(71) In addition, according to an exemplary embodiment to be described below, the display panel **101** further includes a sixth pattern. For example, a third insulating interlayer **200** to be described below is further disposed on the fourth pattern **190** and the sixth pattern is disposed between the third insulating interlayer **200** and the planarization layer **220**, but exemplary embodiments are not limited thereto. The third insulating interlayer **200** and the sixth pattern on the third insulating interlayer **200** may be disposed between the first pattern **130** and the gate insulating layer **140**, between the second pattern **150** and the first insulating interlayer **160**, or between the third pattern **170** and the second insulating interlayer **180**. The sixth pattern includes a scan line, a high potential power line, a bypass line, etc., which will be described below.
(72) According to an embodiment, the third insulating interlayer **200** includes a material substantially the same as a material included in the gate insulating layer **140**, the first insulating interlayer **160**, or the second insulating interlayer **180**.
(73) According to an embodiment, components included in a same pattern all include a substantially same material and are disposed on substantially a same layer. Components in different patterns are disposed on different layers. For example, the semiconductor layer **130** included in the

first pattern **130** is disposed on a different layer from a layer on which the light emission control line EL in the second pattern **150** is disposed. For example, the light emission control line EL is disposed on a layer closer to the pixel electrode PE than a layer on which the semiconductor layer **130** is disposed.
(74) FIG. **6** is a plan view that illustrates a display device according to an exemplary embodiment.
(75) Referring to FIG. **6**, the display panel **101** of the display device **100** according to an exemplary embodiment includes the substrate **110** with a display area **310** and a non-display area **320** at an edge of the display area **310**. The display area **310** is an area in which an input image is displayed and includes pixels PX in a matrix form, which are partitioned by scan lines and data lines that cross each other. In the non-display area **320**, a plurality of driving elements for transmitting driving signals to the display area **310** are disposed. For example, the non-display area **320** includes the scan driver **102**, the data driver **104** and various lines connected to the pixels
(76) PX for driving the pixels PX in the display area **310**.
(77) As illustrated in FIG. **6**, according to an embodiment, the non-display area **320** that surrounds the display area **310** is disposed at the edge of a front surface and is covered by a bezel. Alternatively, the display area **310** can be expanded to the entirety of the front surface of the display device or to one or more side surfaces of the display device, and thus there may be no non-display area **320** or bezel on the front surface. In addition, no pixels PX are disposed in the non-display area **320**. However, exemplary embodiments are not limited thereto and there may be pixels PX disposed in the non-display area **320** as well.
(78) In an exemplary embodiment, at least one sensor is disposed that overlaps the display area **310** of the substrate **110**. The sensor area **330** is an area of the display area **310** that overlaps the sensor. The sensor area **330** is located in the display area **310**. The sensor area **330** is surrounded by the display area **310** and the sensor area **330** is located at one side of the display area **310** as well. In FIG. **6**, the sensor area **330** is depicted as having a substantially circular shape, but exemplary embodiments are not limited thereto. The sensor area **330** may have a polygonal shape, an elliptical shape, a linear shape, etc.
(79) In an exemplary embodiment, the sensor is at least one of a proximity sensor, an illumination sensor, an RGB sensor, an infrared sensor, a finger scan sensor, an ultrasonic sensor, an optical sensor, such as a camera, a microphone, an environmental sensor, such as a barometer, a hygrometer, a thermometer, a radiation sensor, a heat sensor, a gas sensor, etc., or a chemical sensor, such as an electronic nose, a healthcare sensor, a biometric sensor, etc.
(80) According to an exemplary embodiment, the sensor area **330** transmits a signal, such as sound or light, to the sensor. An open hole, referred to hereinafter as a penetrating sensor area, is provided which physically penetrates through the display panel **101** or the substrate **110** at the sensor area **330** so that the signal can pass therethrough. In such an exemplary embodiment, as will be described below, lines, such as scan lines, data lines or power lines, which cross the sensor area **330** are divided into a plurality of line groups and a bypass line is provided that bypasses the sensor area **330** to connect the divided line groups.
(81) According to another exemplary embodiment, the display panel **101** has a non-penetrating sensor area at the sensor area **330** that does not physically penetrate the display panel **101** or substrate **110** and through which a signal can be transmitted. In such an exemplar exemplary embodiment, the scan lines, data lines, and power lines, etc., cross the sensor area without being divided into groups, thereby substantially minimizing interference with signals propagating through the sensor area **330**.
(82) Although exemplary embodiments of the present disclosure have been described with respect to a non-penetrating sensor area, exemplary embodiments can incorporate a penetrating sensor area unless otherwise specified.
(83) In an exemplary embodiment, a wiring connection area **340** is located between the sensor area **330** and the display area **310**. Lines that extend from the display area **310** and lines that extend

from the sensor area **330** are connected to each other at the wiring connection area **340**. For example, two lines can be connected to each other in a straight line, two lines that extend in different directions can be connected to each other, two parallel lines are connected to each other using another line which crosses the two parallel lines, and two lines disposed in different layers with an insulating layer therebetween are connected to each other through a contact hole in the insulating layer. As used herein, one line can be understood to include a plurality of lines connected to each other to transmit the same signal or the same power voltage.
(84) In addition, in an exemplary embodiment, a dummy pixel can be disposed at the wiring connection area **340** between the sensor area **330** and the display area **310**. More specifically, the wiring connection area **340** includes an area **341** adjacent to the sensor area **330** and an area **342** adjacent to the display area **310**, and the dummy pixel is disposed in the area **342** adjacent to the display area **310**. The dummy pixel increases the uniformity of the substrate and substantially prevents damage to a pixel PX adjacent to the sensor area **330**.
(85) FIG. **7** is an enlarged view of an area **351** of FIG. **6** where a wiring connection area meets a sensor area in an X-axis direction according to an exemplary embodiment, and FIGS. **8**, **9** and **10** are cross-sectional views each taken along line I-I′ of FIG. **7**. More specifically, FIG. **7** is a plan view common to exemplary embodiments described below and FIGS. **8**, **9** and **10** are cross-sectional views that respectively illustrate exemplary embodiments.
(86) As illustrated in FIG. **7**, according to an exemplary embodiment, four X-axis direction lines **151**, **152**, **153** and **154** and four Y-axis direction lines **191**, **192**, **193** and **194** extend from the display area to cross a sensor area **330**. The X-axis direction lines **151**, **152**, **153** and **154** illustrated in FIG. **7** may be scan lines SLn−1, SLn and SLn+1 and a light emission control line EL, respectively. Hereinafter, the scan lines SL will include the light emission control line EL. In addition, the Y-axis direction lines **191** and **193** may be data lines DL and the Y-axis direction lines **192** and **194** may be high potential lines VDL. However, exemplary embodiments are not limited thereto and embodiments can include any plurality of lines arranged parallel to each other in substantially a same layer in the display area **310** and that extend into the sensor area **330**.
(87) Hereinafter, according to an exemplary embodiment, it is assumed that four scan lines **151**, **152**, **153** and **154** that extend in the X-axis direction and two data lines **191** and **193** and two high potential lines **192** and **194** that extend in the Y-axis direction extend from the display area **310** into the wiring connection area **340**. In addition, a display device **100** according to an exemplary embodiment includes two auxiliary scan lines. As illustrated in FIG. **7**, two scan lines **151** and **153** of the four scan lines **151**, **152**, **153** and **154** overlap two auxiliary scan lines **211** and **212**, respectively, with an insulating layer therebetween. Hereinafter, such exemplary embodiments will be described in detail.
(88) Hereinafter, an exemplary embodiment will be described more specifically with reference to FIGS. **7** and **8**. Herein, detailed descriptions of components of the display area that have been described above will be omitted.
(89) In an exemplary embodiment, a substrate **110**, a buffer layer **120** and a gate insulating layer **140** are disposed at a wiring connection area **340** and a sensor area **330** in a display area **310**. However, there may be no semiconductor layer **130**, gate electrodes GE**1**, GE**2**, GE**3**, GE**4**, GE**5**, GE**6** and GE**7**, etc formed in the wiring connection area **340** and the sensor area **330**.
(90) In an exemplary embodiment, at least a part **151** and **153** of scan lines **151**, **152**, **153** and **154** extend from the display area **310** into the sensor area **330** in the X-axis direction. On the other hand, another part **152** and **154** of the scan lines **151**, **152**, **153** and **154** extend from the display area **310** into the wiring connection area **340** but do not extend into the sensor area **330**.
(91) In an exemplary embodiment, a first insulating interlayer **160** is disposed on the scan lines **151**, **152**, **153** and **154** and an initialization lines (IL) **172** is disposed on the first insulating interlayer **160**, as in the display area **310**. However, no capacitor electrode **171** is disposed at the sensor area **330**.

(92) According to an exemplary embodiment, a third insulating interlayer **200** is disposed on the initialization line **172**, and contact holes **215** and **216** are formed in the first insulating interlayer **160** and the third insulating interlayer **200** to expose the scan lines **152** and **154** in the wiring connection area **340**.
(93) According to an exemplary embodiment, auxiliary scan lines **211** and **212** are disposed on the third insulating interlayer **200**. The auxiliary scan lines **211** and **212** are disposed in the sensor area **330** and the wiring connection area **340** and are not disposed in the display area **310**. In addition, at least a part of the auxiliary scan lines **211** and **212** overlap the scan lines **151** and **153**, respectively, in the sensor area **330** in a plan view. The auxiliary scan lines **211** and **212** respectively include bent portions **213** and **214** in the wiring connection area **340** and are connected to the scan lines **152** and **154** through the respective contact holes **215** and **216** formed in the first insulating interlayer **160** and the third insulating interlayer **200**.
(94) Next, according to an exemplary embodiment, a second insulating interlayer **180** is disposed on the auxiliary scan lines **211** and **212**. In addition, as in the display area **310**, data lines **191** and **193** and high potential lines **192** and **194** are disposed on the second insulating interlayer **180**. A planarization layer **220** and a sealing member **270** are disposed on the data lines **191** and **193** and the high potential lines **192** and **194**.
(95) According to an exemplary embodiment, the third insulating interlayer **200** includes a material substantially the same as a material included in the second insulating interlayer **180**. The auxiliary scan lines **211** and **212** include a material substantially the same as a material included in the scan line SL or the data line DL.
(96) In an exemplary embodiment, a pixel electrode PE, a light blocking layer **240**, a light emitting layer **250**, a common electrode **260**, etc., are not formed in the sensor area **330** or are removed therefrom. For example, the light emitting layer **250** and the common electrode **260** can be formed with the display area **310** and then removed using a laser energy source. In addition, the common electrode **260** can be deposited through a metal self patterning method such that the common electrode **260** is formed in the display area **310** but not in the sensor area **330**. Such a structure is not limited to an exemplary embodiment and can be part of all exemplary embodiments described below.
(97) In addition, according to an exemplary embodiment, at least one of insulating layers such as the buffer layer **120**, the gate insulating layer **140**, the first, second and third insulating interlayers **160**, **180** and **200** or the planarization layer **220** is not formed in or is removed from an area that does not overlap the above-described lines **151**, **153**, **211**, **212**, **191** and **194** in the sensor area **330**. For example, after the light emitting layer **220** and the common electrode **260** are removed as described above, the gate insulating layer **140**, the first, second and third insulating interlayers **160**, **180** and **200** and the planarization layer **220** are removed. Thereafter, a sealing member **270** is disposed. No insulating layer is disposed on the area that does not overlap any of the above-described lines **151**, **153**, **211**, **212**, **191** and **194** in the sensor area **330**. However, to ensure insulation of the above-described lines **151**, **153**, **211**, **212**, **191**, **192**, **193** and **194**, an insulating layer remains in an area adjacent to the above-described lines without being removed. The insulating layer removal structure is not only limited to an exemplary embodiment, but may part of all exemplary embodiments described below.
(98) Hereinafter, an exemplary embodiment of the present disclosure will be described more specifically with reference to FIGS. **7** and **9**. Herein, descriptions of components described hereinabove with respect to FIGS. **7** and **8** will be omitted.
(99) According to an exemplary embodiment, a substrate **110**, a buffer layer **120**, a gate insulating layer **140**, scan lines **151**, **152**, **153** and **154**, a first insulating interlayer **160** and an initialization line **172** are disposed as shown in FIG. **8**.
(100) According to an exemplary embodiment, a second insulating interlayer **180** is disposed on the initialization line **172**. In addition, data lines **191** and **193** and high potential lines **192** and **194** are

disposed on the second insulating interlayer **180**.

(101) According to an exemplary embodiment, a third insulating interlayer **200** is disposed on the data lines **191** and **193** and the high potential lines **192** and **194**. Contact holes **215** and **216** are formed in the first insulating interlayer **160**, the second insulating interlayer **180**, and the third insulating interlayer **200** to expose the scan lines **152** and **154** at the wiring connection area **340**.

(102) In addition, according to an exemplary embodiment, auxiliary scan lines **211** and **212** are disposed on the third insulating interlayer **200**. At least a part of the auxiliary scan lines **211** and **212** overlap the scan lines **151** and **153**, respectively, in the sensor area **330** in a plan view. The auxiliary scan lines **211** and **212** respectively include bent portions **213** and **214** in the wiring connection area **340** and are connected to the scan lines **152** and **154** through the respective contact holes **215** and **215** formed in the first insulating interlayer **160**, the second insulating interlayer **180** and the third insulating interlayer **200**.

(103) Next, according to an exemplary embodiment, a planarization layer **220** and a sealing member **270** are disposed on the auxiliary scan lines **211** and **212**.

(104) Hereinafter, an exemplary embodiment of the present disclosure will be described more specifically with reference to FIGS. **7** and **10**. Herein, descriptions of components described hereinabove with respect to FIGS. **8** and **9** will be omitted.

(105) According to an exemplary embodiment, a substrate **110**, a buffer layer **120**, a gate insulating layer **140**, scan lines **151**, **152**, **153** and **154**, a first insulating interlayer **160** and an initialization line **172** are disposed as shown in FIG. **9**.

(106) According to an exemplary embodiment, a second insulating interlayer **180** is disposed on the initialization line **172**. In addition, data lines **191** and **193** and high potential lines **192** and **194** are disposed on the second insulating interlayer **180**. In addition, a planarization layer **220** is disposed on the data lines **191** and **193** and high potential lines **192** and **194**.

(107) According to an exemplary embodiment, contact holes **215** and **216** are formed in the first insulating interlayer **160**, the second insulating interlayer **180**, and the planarization layer **220** to expose the scan lines **152** and **154** in the wiring connection area **340**. Auxiliary scan lines **211** and **212** are disposed on the planarization layer **220** in the sensor area **330** and the wiring connection area **340**. At least a part of the auxiliary scan lines **211** and **212** overlap the scan lines **151** and **153**, respectively, in the sensor area **330** in a plan view. The auxiliary scan lines **211** and **212** respectively include bent portions **213** and **214** in the wiring connection area **340** and are connected to the scan lines **152** and **154** through the respective contact holes **215** and **216** formed in the first insulating interlayer **160**, the second insulating interlayer **180**, and the planarization layer **220**.

(108) Next, according to an exemplary embodiment, a sealing member **270** is disposed on the auxiliary scan lines **211** and **212**. The auxiliary scan lines **211** and **212** and a pixel electrode PE are formed substantially simultaneously using substantially a same material.

(109) In the above-described exemplary embodiments, the scan lines **151** and **153** extend from the display area **310** into the sensor area **330** in the X-axis direction in substantially a same layer, and the auxiliary scan lines **211** and **212** overlap the scan lines **151** and **153** in the sensor area **330**, but exemplary embodiments are not limited thereto. For example, the scan lines **151** and **154** may extend from the display area **310** into the sensor area **330** in the X-axis direction on a substantially same layer and the auxiliary scan lines **211** and **212** may overlap the scan lines **151** and **154** in the sensor area **330**.

(110) In addition, according to an exemplary embodiment, although FIG. **7** shows that the widths of the scan lines **151** and **153** differs from the widths of the auxiliary scan lines **211** and **212**, the widths of the scan lines **151** and **153** and the auxiliary scan lines **211** and **212** may be substantially equal to each other. In addition, the scan lines **151** and **153** and the auxiliary scan lines **211** and **212** may partially overlap each other in the width direction.

(111) According to exemplary embodiments, as an area occupied by the scan lines **151** and **153** and the auxiliary scan lines **211** and **212** in the sensor area **330** in a plan view is reduced by about ½

and an interval between the scan lines **151** and **153** and the auxiliary scan lines **211** and **212** increases, signal transmittance in the sensor area **330** can be increased.
(112) In addition, according to exemplary embodiments, although the scan line **151** and the auxiliary scan line **211** overlap each other in the sensor area **330**, exemplary embodiments are not limited thereto. For example, three or more lines may overlap each other on different layers in substantially the same manner as in above exemplary embodiments.
(113) In exemplary embodiments, although the scan lines **151** and **153** and the auxiliary scan lines **211** and **212** overlap each other at the sensor area **330**, exemplary embodiments are not limited thereto. For example, the data lines **191** and **193** and power lines **192** and **194** may overlap each other in the sensor area in the same manner as in above-described exemplary embodiments.
(114) Hereinafter, an exemplary embodiment of the present disclosure will be described more specifically with reference to FIGS. **7** and **11**. Herein, detailed descriptions of the components of the display area described above with respect to FIGS. **8-10** will be omitted.
(115) FIG. **11** is a cross-sectional view taken along line of FIG. **7**. For ease of description, a floating data line DL and high potential line VDL is described based on a stack structure of FIG. **8**. However, exemplary embodiments are not limited thereto and the floating line may be described based on the stack structures of FIG. **9** or **10** or other exemplary embodiments described hereinbelow.
(116) According to exemplary embodiments, data lines **191** and **193** and high potential lines **192** and **194** are disposed on a second insulating interlayer **180** as depicted in FIG. **11** and described with respect to FIG. **8**. A planarization layer **220** and a sealing member **270** are disposed on the data lines **191** and **193** and the high potential lines **192** and **194**.
(117) In addition, according to exemplary embodiments, a first floating line **155** is disposed between a gate insulating layer **140** and a first insulating interlayer **160** and a second floating line **218** is disposed between a third insulating interlayer **200** and the second insulating interlayer **180**. In addition, a third floating line may be disposed between the first insulating interlayer **160** and the third insulating interlayer **200**. The data line **191**, the first floating line **155** and the second floating line **218** are connected to each other through contact holes **195** and **196** formed in the first insulating interlayer **160**, the third insulating interlayer **200** and the second insulating interlayer **180**. The first floating line **155**, the second floating line **218** and the third floating line overlap the data line **191** in a plan view. Each of the first floating line **155**, the second floating line **218** and the third floating line have a width substantially equal to a width of the data line **191** and are not long enough to contact the scan lines **151** and **153**, the auxiliary scan lines **211** and **212** and an initialization line **172**.
(118) According to exemplary embodiments, the first floating line **155** and the scan lines SL on the gate insulating layer **140** are formed substantially simultaneously using substantially a same material, the second floating line **218** and the auxiliary scan lines **211** and **212** on the third insulating interlayer **200** are formed substantially simultaneously using a substantially same material, and the third floating line and the initialization lines IL on the first insulating interlayer **160** are formed substantially simultaneously using substantially a same material.
(119) According to exemplary embodiments, the floating lines **155** and **218** formed to reduce a resistance of the data line **191**. Hereinabove, although an embodiment illustrated in FIG. **11** shows, for example, the floating lines **155** and **218** overlapping the data line **191**, exemplary embodiments are not limited thereto. Exemplary embodiments may include a power line such as the high potential lines **192** and **194**, the scan lines **151**, **152**, **153** and **154** and the auxiliary scan lines **211** and **212**.
(120) FIG. **12** is an enlarged view of an area **351** of FIG. **6** where a wiring connection area meets the sensor area in the X-axis direction according to an exemplary embodiment and FIG. **13** is a cross-sectional view taken along line I-I′ of FIG. **12**.
(121) Hereinafter, an exemplary embodiment of the present disclosure will be described more

specifically with reference to FIGS. **12** and **13**. Herein, detailed descriptions of components of a display area described hereinabove will be omitted.

(122) As illustrated in FIGS. **12** and **13**, an exemplary embodiment f the present disclosure is directed to a rerouting of an initialization line IL that is disposed on a different layer from that on which scan lines **151**, **152**, **153** and **154** and auxiliary scan lines **211** and **212** are disposed. Hereinafter, for ease of description, descriptions will be given based on an exemplary embodiment shown in FIG. **8**.

(123) According to exemplary embodiments, as described hereinabove for FIG. **8**, a buffer layer **120**, a gate insulating layer **140**, the scan lines **151**, **152**, **153** and **154**, the auxiliary scan lines **211** and **212**, initialization lines **172** and **173**, data lines **191** and **193** and high potential lines **192** and **193** are disposed on a substrate **110**. The auxiliary scan lines **211** and **212** overlap the scan lines **151** and **153**, respectively, in a sensor area **330** in a plan view.

(124) According to exemplary embodiments, the initialization lines **172** and **173** are disposed between the first insulation layer **160** and the third insulation layer **200** in areas **310**, **330** and **340**. That is, the initialization lines **172** and **173** are disposed in a same layer that is a different layer from that on which the scan lines **151**, **152**, **153** and **154** and the auxiliary scan lines **211** and **212** are disposed.

(125) According to exemplary embodiments, in the display area **310**, the initialization line **172** is disposed between the scan lines **151**, **152**, **153** and **154** and thus do not overlap the scan lines **151**, **152**, **153** and **154** in a plan view. In the sensor area **330**, the initialization line **173** overlaps the scan lines **153** and **212**. In the wiring connection area **340**, the initialization line **174** includes bent portions **175** and **176** that connect the initialization line **172** to the initialization line **173**.

(126) According to an exemplary embodiment, as the initialization line **173** overlaps the scan lines in the sensor area **330**, an area of the sensor area **330** can be reduced. Accordingly, an interval between the lines in the sensor area **330** can be increased.

(127) Although an exemplary embodiment has been described as compared to an exemplary embodiment shown in FIG. **8**, similar features can be incorporated into exemplary embodiments shown in FIGS. **9-10**. In addition, in an exemplary embodiment, the initialization lines overlap any one of the scan lines **151**, **152** and **154** instead of overlapping scan lines **153** and **212**. Although an exemplary embodiment has been described with respect to the initialization line IL by way of example, exemplary embodiments are not limited thereto, and other lines may be rerouted to overlap the scan lines.

(128) FIG. **14** is a wiring diagram of a high potential line VDL around sensor area **330** according to an exemplary embodiment. Although FIG. **14** depicts a high potential line VDL by way of example, exemplary embodiments are not limited thereto. For example, exemplary embodiments may incorporate other kinds of power lines, e.g., a low potential line VSL and an initialization line IL. In addition, exemplary embodiments may incorporate not only a non-penetrating sensor area but also a penetrating sensor area.

(129) According to exemplary embodiments, as illustrated in FIG. **14**, the high potential line VDL includes a plurality of lines **195** and **196**, hereinafter referred to as “main power lines”, in a first direction, such as the Y-axis direction, and a plurality of lines **178**, hereinafter referred to as “auxiliary power lines”, in a second direction, such as the X-axis direction, which crosses the first direction. The main power lines **195** and **196** and the auxiliary power line **178** are connected to each other at crossing points **197** and **198** to form a mesh. In addition, the high potential lines VDL, i.e., the main power lines **195** and **196** and the auxiliary power line **178**, are not disposed in the sensor area **330**.

(130) According to exemplary embodiments, the main power lines **195** and **196** and the auxiliary power line **178** are disposed in substantially a same layer. The main power lines **195** and **196** and the auxiliary power line **178** are disposed in a different layer from that on which the scan line SL and the data line DL are disposed. In addition, the main power lines **195** and **196** and the auxiliary

power line **178** are formed on the third insulating interlayer **200**. Since auxiliary scan lines **211** and **212** are disposed in the sensor area **330** but not in the display area **310**, the auxiliary scan lines **211** and **212** and the high potential lines, i.e. the main power lines **195** and **196** and the auxiliary power line **178**, do not contact each other and are formed substantially simultaneously on the third insulating interlayer **200**.
(131) Alternatively, according to exemplary embodiments, the main power lines **195** and **196** are disposed on a different layer from that on which the auxiliary power line **178** is disposed, as described below in which the main power lines **195** and **196** and the auxiliary power line **178** are connected to each other through contact holes formed at the crossing points **197** and **198**.
(132) FIG. **15** is an enlarged view of an area **352** in FIG. **6** at which a sensor area meets a display area in a Y-axis direction, according to an exemplary embodiment and FIG. **16** is a cross-sectional view taken along the line I-I′ of FIG. **15**. Herein, detailed descriptions of components of a display area described hereinabove will be omitted. Although an exemplary embodiment is described FIG. **7** for ease of description, exemplary embodiments are not limited thereto. Exemplary embodiments may incorporate a structure shown in FIG. **9**, or may structures other than those shown in FIGS. **8-9**.
(133) According to exemplary embodiments, as illustrated in FIGS. **14** and **15**, high potential lines **195**, **196** and **178** are not disposed in a sensor area **330**. That is, main power lines **195** and **196** extend from a display area **310** in a Y-direction extend only into a wiring connection area **340**, or do not extend past the display area **310**). In addition, auxiliary power lines extend from the display area **310** in a X-direction extend into the wiring connection area **340**, and do not extend into the sensor area **330**.
(134) Referring to FIGS. **15** and **16**, according to exemplary embodiments, a a buffer layer **120** and a gate insulating layer **140** are disposed on substrate **110**, as shown in FIG. **8**. A first insulating interlayer **160** is disposed on the scan lines **151**, **152**, **153** and **154** and an initialization line (IL) **172** is disposed on the first insulating interlayer **160**.
(135) In addition, according to exemplary embodiments, an auxiliary power line **178** is formed along with the initialization line **172**. That is, the auxiliary power line **178** is disposed between the initialization lines **172** on substantially a same layer as the initialization line **172**. As described above, the auxiliary power line **178** is not disposed in the sensor area **330**. A third insulating interlayer **200** is disposed on the initialization line **172** and the auxiliary power line **178**, and auxiliary scan lines **211** and **212** are disposed on the third insulating interlayer **200**.
(136) Next, according to exemplary embodiments, a second insulating interlayer **180** is disposed on the auxiliary scan lines **211** and **212**. A contact hole **197** is formed in the third insulating interlayer **200** and the second insulating interlayer **180** to expose the auxiliary power line **178** therethrough.
(137) Next, according to exemplary embodiments, data lines **191** and **193** and the main power lines **195** and **196** are disposed on the second insulating interlayer **180**. That is, the main power lines **195** and **196** are disposed between the data lines **191** and **193** on substantially a same layer as the layer on which the data lines **191** and **193** are disposed. In addition, as described above, the main power lines **195** and **196** are not disposed in the sensor area **330**. The main power lines **195** and **196** are connected to the auxiliary power line **178** through the contact holes **197** and **198** formed in the third insulating interlayer **200** and the second insulating interlayer **180**. A planarization layer **220** and a sealing member **270** are disposed on the data lines **191** and **193** and the main power lines **195** and **196**.
(138) Hereinabove, according to exemplary embodiments, although the main power lines **195** and **196** are described as being disposed on substantially a same layer as the data lines **191** and **193**, and the auxiliary power line **178** is disposed on substantially a same layer as the initialization line **172**, exemplary embodiments are not limited thereto. For example, the main power lines **196** and **196** may be disposed on a layer on which Y-axis direction lines are disposed and the auxiliary power line **178** may be disposed on a layer on which X-axis direction lines are disposed.

(139) FIG. **17** is an enlarged view of an area a**352** of FIG. **6** where a sensor area meets a display area in a Y-axis direction, according to an exemplary embodiment and FIG. **18** is a cross-sectional view taken along the line I-I′ of FIG. **17**. Hereinafter, an exemplary embodiment will be described with reference to FIGS. **14**, **17** and **18**. Herein, detailed descriptions of components of the display area described hereinabove will be omitted. As described hereinabove with respect to FIGS. **15-16**, high potential lines **195**, **196** and **178** are not disposed at the sensor area **330**.
(140) Referring to FIGS. **17** and **18**, according to exemplary embodiments, a buffer layer **120** and a gate insulating layer **140** are disposed on substrate **110**, as shown in FIG. **16**. A first insulating interlayer **160** is disposed on the scan lines **151**, **152**, **153** and **154**, and an initialization line (IL) **172** is disposed on the first insulating interlayer **160**. In addition, an auxiliary power line **178** is formed along with the initialization line **172**.
(141) According to exemplary embodiments, a second insulating interlayer **180** is disposed on the initialization line **172** and the auxiliary power line **178**, and auxiliary scan lines **211** and **212** are disposed on the second insulating interlayer **180**. In addition, data lines **191** and **193** and main power lines **195** and **196** are disposed on the second insulating interlayer **180**, but the data lines **191** and **193** and the main power lines **195** and **196** are not disposed in the sensor area **330**. That is, the main power lines **195** and **196** extend from the display area **310** in a Y-direction extend only into a wiring connection area **340**.
(142) In addition, according to exemplary embodiments, the main power lines **195** and **196** are connected to an auxiliary power line **178** through a contact hole **197** formed in the second insulating interlayer **180**. A planarization layer **220** is disposed on the data lines **191** and **193** and the main power lines **195** and **196** and contact holes **233** and **234** are formed in the planarization layer **220** to expose the data lines **191** and **193** in the wiring connection area **340**.
(143) According to exemplary embodiments, the data lines **231** and **232** are disposed on the planarization layer **220** in the sensor area **330** and the wiring connection area **340**. The data lines **231** and **232** and the pixel electrode PE are formed substantially simultaneously using substantially a same material. Opposite end portions of the data lines **231** and **232** are disposed, one at one wiring connection area **340** and the other at an opposite wiring connection area **340**. Data lines **231** and **232** are connected to data lines **191** and **192** through contact holes **233** and **234** formed in the planarization layer **220**. A sealing member **270** is disposed on data lines **231** and **232**.
(144) FIGS. **19** and **20** are wiring diagrams of display devices according to exemplary embodiments. Hereinafter, a bypass line that bypasses a sensor area will be described in detail with reference to FIGS. **19** and **20**.
(145) According to exemplary embodiments, FIGS. **19** and **20** illustrate an upper end portion of the display device **100** that includes a plurality of sensor areas of FIG. **6**. In the following description, horizontal, vertical, left, right, upper and lower directions are based on a display device of FIG. **6**. As described hereinabove with reference to FIG. **1**, a display device **100** according to an exemplary embodiment includes a display panel **101**, a scan driver **102** and a data driver **103**.
(146) As described above according to exemplary embodiments, the display panel **101** includes a plurality of sensor areas **331**, **332** and **333** in a display area **310**. Accordingly, as illustrated in FIG. **19**, scan lines that extend in a X-direction and that cross the plurality of sensor areas **331** and **332** are divided into three groups of lines by the plurality of sensor areas **3331** and **332**. That is, scan lines that extend in a X-direction and that cross the sensor areas **331** and **332** include first line group **430** that includes lines **431**, **432**, **433** and **434** that have one end directly connected to the scan driver **102** and another end cut by the sensor area **331**, second line group **440** that includes lines **441**, **442**, **443** and **444** with opposite ends respectively cut by the sensor areas **331** and **332**, and third line **450** that includes line **451**, **452**, **453** and **454** that have one end cut by the sensor area **332** and another end that extends toward the opposite side of the scan driver **102**. On the other hand, the other scan lines **435**, **436**, **437** and **438** are not separated by the sensor area **330**.
(147) Referring to FIG. **19**, according to exemplary embodiments, bypass line group includes lines

461, **462**, **463** and **464** that connect lines **431**, **432**, **433** and **434** of the first line group **430** and lines **441**, **442**, **443** and **444** of the second line group **440**, respectively, and bypass line group **465** includes lines **466**, **467**, **468** and **469** that connect lines **441**, **442**, **443** and **444** of the second line group **440** and lines **451**, **452**, **453** and **454** of the third line group **450**, respectively. Bypass lines **461**, **462**, **463** and **464** are spaced apart from the scan lines **431**, **432**, **433**, **434**, **435**, **436**, **437** and **438** with an insulating layer, such as a third insulating interlayer **200**, therebetween. Bypass lines **461**, **462**, **463** and **464** are connected to the lines **431**, **432**, **433** and **434**, respectively, through contact hole group **470** that includes contact holes **471**, **472**, **473** and **474** formed in the insulating layer, and are connected to the lines **441**, **442**, **443** and **444**, respectively, through contact holes **481**, **482**, **483** and **484** of contact hole group **480** formed in the insulating layer. Accordingly, the lines **431**, **432**, **433** and **434** of the first line group **430** are connected to the lines **441**, **442**, **443** and **444** or the second line group **440**, respectively, through bypass lines **461**, **462**, **463** and **464**, such that a scan signal is transmitted to the lines **441**, **442**, **443** and **444**.

(148) Similarly, according to exemplary embodiments, other bypass lines **466**, **467**, **468** and **469** of bypass line group **465** connect the lines **441**, **442**, **443** and **444** of the second line group **440** and the lines **451**, **452**, **453** and **454** of the third line group **450**, respectively. Accordingly, the lines **441**, **442**, **443** and **444** are connected to the lines **451**, **452**, **453** and **454** through the bypass lines **466**, **467**, **468** and **469** of bypass line group **465**, respectively, such that a scan signal is transmitted to the lines **451**, **452**, **453** and **454**.

(149) For example, according to exemplary embodiments, the bypass lines **461**, **462**, **463** and **464** extend in the Y-axis direction in the display area **310** from the contact holes of contact hole group **470** in the display area **310** on a left side of the sensor area **331**, extend in the X-axis direction in the display area **310** to an upper side of the sensor area **331**, and extend in the Y-axis direction to contact holes **481**, **482**, **483** and **484** of contact hole group **480** located in the display area **310** on a right side of the sensor area **331**.

(150) Similarly, according to exemplary embodiments, other bypass lines **466**, **467**, **468** and **469** of bypass line group **465** extend in the Y-axis direction in the display area **310** from contact holes **485**, **486**, **487** and **488** of contact hole group **480** located in the display area **310** on a left side of a sensor area **332**, extend in the X-axis direction in the display area **310** to an upper side of the sensor area **332**, and extend in the Y-axis direction to contact holes **491**, **492**, **493** and **494** of contact hole group **490** located in the display area **310** on a right side of the sensor area **312**.

(151) Although bypass line groups **460** and **465** are described as extending in the X-axis direction and the Y-axis directions in the display area **310**, exemplary embodiments are not limited thereto. The contact holes **470** and **490** may be located in the non-display area **320** at an edge of the display area **310** and the bypass lines **460** and **465** may extend along the non-display area **320** in the X-axis direction or the Y-axis directions. However, the contact holes of contact hole group **480** for connecting lines of second line group **440** between the plurality of sensor areas **331** and **332** are located at the display area **310**.

(152) In addition, although the bypass line groups **460** and **465** are described as extending in the X-axis direction and the Y-axis directions, exemplary embodiments are not limited thereto. For example, the bypass lines **460** and **465** have an oblique straight shape or a curved line shape. In addition, the bypass lines **460** and **465** may be curved in a shape similar to that of the sensor areas **330**.

(153) In addition, although the lines **431**, **432**, **433** and **434** of first line group **430** and the lines of second line group **440** are connected to each other by the line of bypass line group **460**, and the lines of the second line group **440** and the lines of the third line group **450** are connected to each other by lines of bypass line group **465**, exemplary embodiments are not limited thereto. For example, the lines of the first line group **430** and the second line group **440** may be connected to each other by the lines of bypass line group **460** and the lines **431**, **432**, **433** and **434** of first line group **430** and the lines of the third line group **450** may be connected to each other by other bypass

lines.

(154) In an exemplary embodiment, lines of the bypass line groups **460** and **465** are disposed between the third insulating interlayer **200** and the second insulating interlayer **180** as shown in FIG. **8**, and the lines of bypass line groups **460** and **465** and each of the line groups **430**, **440** and **450** are connected to each other through the holes of contact hole groups **470**, **480** and **490** formed in the first insulating interlayer **160** and the third insulating interlayer **200**.

(155) FIG. **20** is a bypass line diagram according to an exemplary embodiment.

(156) Referring to FIG. **20**, bypass lines include bypass line group **540** with bypass lines **541**, **542**, **543**, and **544** that extend in the X-axis direction parallel to scan lines SL and bypass line group **510** with bypass lines **511**, **512**, **523**, and **514**, bypass line group **520** with bypass lines **521**, **522**, **523**, and **524**, and bypass group **530** with bypass lines **531**, **532**, **533**, and **534**, all of which extend in the Y-axis direction. The bypass lines of bypass group **540** extending in the X-axis direction are formed on a pattern layer on which lines in the X-axis direction, such as scan lines SLn and EL and initialization line IL, are formed. The bypass lines of bypass lines groups **510**, **520** and **530** extending in the Y-axis direction are formed on a pattern layer on which lines in the Y-axis direction, such as data lines DL, are formed. Accordingly, the bypass lines of bypass line group **540** extending in the X-axis direction and the bypass lines of bypass line groups **510**, **520** and **530** extending in the Y-axis direction are spaced apart from each other with a first insulating interlayer and a second insulating interlayer, and in some embodiments, a third insulating interlayer, therebetween, and are connected to each other through contact holes formed in the insulating layers.

(157) More specifically, according to exemplary embodiments, the bypass lines of bypass line group **540** extending in the X-axis direction extend along a non-display area **320**. The bypass lines of bypass line group **510** extending in the Y-axis direction have one end connected to lines **431**, **432**, **433** and **434** of line group **430**, through contact holes **516**, **517**, **518** and **519**, respectively, of contact hole group **515** located in the non-display area **320** and another end connected to the bypass lines of bypass line group **540** extending in the X-axis direction through contact holes located at the non-display area **320**. The bypass lines of bypass line group **520** extending in the Y-axis direction have one end connected to lines of the second line group **440** through a contact holes **526**, **527**, **528** and **529** of bypass line group **525** located in the display area **310** and another end connected to the bypass lines of bypass line group **540** extending in the X-axis direction through contact holes located in the non-display area **320**. The bypass lines of bypass line group **530** extending in the Y-axis direction have one end connected to lines of third line group **450** through contact holes **536**, **537**, **538** and **539** of contact hole group **535** located in the non-display area **320** and another end connected to the bypass lines of bypass line group **540** extending in the X-axis direction through contact holes located at the non-display area **320**. Accordingly, the bypass lines of bypass line groups **515** and **535** extending in the Y-axis direction extend along the non-display area **320** and the bypass lines of bypass line group **520** extending in the Y-axis direction are disposed in the non-display area **320** and the display area **310**.

(158) Accordingly, in exemplary embodiments, the lines **431**, **432**, **433** and **434** of the first line group **430**, the lines of the second line group **440** and the lines of the third line group **450** are connected to each other through lines of the bypass line groups **510**, **520** and **530** extending in the Y-axis direction and the line of bypass line group **540** that extend in the X-axis direction, such that a scan signal is transmitted thereto.

(159) Although the lines of bypass line group **540** have been described as extending in the X-axis direction along the non-display area **320**, exemplary embodiments are not limited thereto. The lines of bypass line group **540** may extend in the X-axis direction in the display area **310**, for example, between the scan lines SL.

(160) In addition, according to exemplary embodiments, although the contact holes of contact hole groups **515** and **535** have been described as being located in the non-display area **320**, exemplary

embodiments are not limited thereto. That is, the contact holes of contact hole groups **515** and **535** may be located in the display area **310** and the bypass lines of bypass line groups **510** and **530** may be disposed in the display area **310**.
(161) In addition, an additional scan driver may be disposed on the opposite side of the scan driver **102** and the additional scan driver may supply scan signals to the lines of the third line group **450**. In addition, the scan driver **102** may be a single scan driver disposed on the upper side or the lower side of the display panel. A single scan driver may be a driver in which the scan driver **102** and the data driver **103** are integrated. The single scan driver on the upper side or the lower side of the display panel transmits scan signals to the scan lines **431**, **432**, **433**, **434**, **435**, **436**, **437** and **438** or the lines of bypass line groups **460**, **465** and **520** through lines extending in the Y-axis direction along the non-display area **320** at a left edge and/or a right edge of the display panel **101** or lines disposed in the Y-axis direction in the display area **310**.
(162) In addition, according to exemplary embodiments, when the bypass wirings are disposed in the display area, to maintain uniformity between portions where the bypass wirings are disposed and portions where the bypass wirings are absent, dummy wirings that have a substantially same shape as the bypass wirings may be disposed at portions where the bypass wirings are absent.
(163) Hereinabove, according to exemplary embodiments, bypass wirings of a display panel that include the plurality of sensor areas **311** and **312** have been described with reference to FIGS. **19** and **20**. Although bypass wirings have been described with respect to the bypass wirings of the scan lines SL, exemplary embodiments are not limited thereto. The data lines DL, like the scan lines and the bypass lines, may be divided into groups of the same type.
(164) FIGS. **21**A, **21**B, **21**C and **21**D are cross-sectional views illustrating a touch panel and a sealing member of a display device according to an exemplary embodiment.
(165) As illustrated in FIGS. **5** and **21**A, according to exemplary embodiments, a display device **100** includes a sealing member **270** above a display panel **101**. The sealing member **270** is formed into a thin film encapsulation structure in which one or more inorganic layers **271** and one or more organic layers **272**, **273** are alternately stacked along a Z-axis direction. In addition, the display device **100** includes a touch panel **290** on the sealing member **270**. The touch panel **290** includes a metal touch electrode **291** that senses a touch and an insulating layer, and adhesive layer or protective films **292** and **293** above or below the touch electrode **291**.
(166) As illustrated in FIG. **21**B, according to exemplary embodiments, the touch electrode **291** in the touch panel **290** is disposed in a display area **310** and a wiring connection area **340**, and the touch electrode **291** is not disposed in a sensor area **330**. That is, the touch electrode **291** is not formed in the sensor area **330** or is removed therefrom.
(167) In addition, according to exemplary embodiments, as illustrated in FIG. **21**C, the sealing member **270** has a structure in which organic layer **271** and inorganic layers **272** and **271** are alternately disposed in the display area **310** and the wiring connection area **340**, and the inorganic layers **272** and **273** are disposed in the sensor area **330**, and the organic layer **271** is not disposed in the sensor area **330**. That is, the organic layer **271** is not formed in the sensor area **330** or is removed therefrom.
(168) In addition, according to exemplary embodiments, as illustrated in FIG. **21**D, the touch panel **290** is disposed in the display area **310** and the wiring connection area **340**, but not in the sensor area **330**. That is, in addition to the touch electrode **291**, the insulating layer and the adhesive layer or protective films **292** and **293** are not formed at the sensor area **330** or are removed therefrom.
(169) In an exemplary embodiment, as shown in FIG. **8**, a pixel electrode PE, a light blocking layer **240**, a light emitting layer **250**, a common electrode **260**, etc., are not formed in the sensor area **330** of the display panel **101**, or are removed therefrom. In addition, at least one of a buffer layer **120**, a gate insulating layer **140**, first, second and third insulating interlayers **160**, **180** and **200** and a planarization layer **220** are not formed at or are removed from an area in the sensor area **330** of the display panel **101** that does not overlap the above-described lines **151**, **153**, **211**, **212**, **191** and **194**.

(170) As set forth hereinabove, in a display device according to one or more exemplary embodiments, an area that is occupied by wirings that traverse a sensor area can be substantially minimized in a plan view. Accordingly, signal transmittance of the sensor area may be improved.
(171) In addition, in a display device according to one or more exemplary embodiments, signal transmittance in the sensor area can be improved by reducing the number of insulating layers disposed in the sensor area.
(172) While embodiments of the present disclosure have been illustrated and described with reference to the exemplary embodiments thereof, it will be apparent to those of ordinary skill in the art that various changes in form and detail may be made thereto without departing from the spirit and scope of the present invention.

Claims

1. A display device comprising: a substrate that includes a display area in which a plurality of pixels are disposed, a first sensor area in the display area wherein the first sensor area overlaps a first sensor; a plurality of scan lines connected to the plurality of pixels and that extend in a first direction, wherein the plurality of scan lines comprises: a first wiring and a second wiring disposed in the display area and that extend in a direction that crosses the first sensor area, wherein the first wiring is separated from the second wiring by the first sensor area, and the first wiring and the second wiring do not bypass the first sensor area; a third wiring that bypasses the first sensor area and is connected to the first wiring and second wiring; and a first insulating layer disposed on the first wiring and second wiring, and below the third wiring, wherein the third wiring is vertically spaced apart from the first wiring and the second wiring with the first insulating layer interposed therebetween, wherein the third wirings comprises: a first portion that extends in a second direction that intersects the first direction and is connected to the first wirings; a second portion that extends in the second direction and is connected to the second wirings; and a third portion that extends in the first direction and is connected to the first portion and the second portion.
2. The display device of claim 1, wherein a contact hole penetrates the first insulating layer through which the first wirings and the second wiring are connected to the third wiring.
3. The display device of claim 1, wherein the first portion and the second portion are disposed in the display area.
4. The display device of claim 1, wherein the third portion is disposed in the display area.
5. The display device of claim 1, wherein the first portion and the second portion are disposed in a non display area at an edge of the display area.
6. The display device of claim 1, wherein the third portion is disposed in a non display area at an edge of the display area.
7. The display device of claim 1, wherein the display device further comprises a plurality of data lines connected to the plurality of pixels and that extend in a second direction that intersects the first direction.
8. The display device of claim 7, at least one of the plurality of data lines is disposed in the first sensor area.
9. A display device comprising: a substrate that includes a display area in which a plurality of pixels are disposed, a first sensor area in the display area wherein the first sensor area overlaps a first sensor; a plurality of scan lines connected to the plurality of pixels and that extend in a first direction, wherein the plurality of scan lines comprises: a first wiring and a second wiring disposed in the display area and that extend in a direction that crosses the first sensor area, wherein the first wiring is separated from the second wiring by the first sensor area, and the first wiring and the second wiring do not bypass the first sensor area; a third wiring that bypasses the first sensor area and is connected to the first wiring and second wiring; and a first insulating layer disposed on the first wiring and second wiring, and below the third wiring, wherein the third wiring is vertically

spaced apart from the first wiring and the second wiring with the first insulating layer interposed therebetween, wherein the third wirings cross at least one of the plurality of scan lines in a plan view, wherein the at least one of the plurality of scan lines is not connected to first, second and third wirings.

10. A display device comprising: a substrate that includes a display area in which a plurality of pixels are disposed, a first sensor area in the display area wherein the first sensor area overlaps a first sensor; a plurality of scan lines connected to the plurality of pixels and that extend in a first direction, wherein the plurality of scan lines comprises: a first wiring and a second wiring disposed in the display area and that extend in a direction that crosses the first sensor area, wherein the first wiring is separated from the second wiring by the first sensor area, and the first wiring and the second wiring do not bypass the first sensor area; a third wiring that bypasses the first sensor area and is connected to the first wiring and second wiring; and a first insulating layer disposed on the first wiring and second wiring, and below the third wiring, wherein the third wiring is vertically spaced apart from the first wiring and the second wiring with the first insulating layer interposed therebetween, wherein the substrate further includes a second sensor area that overlaps a second sensor, and the plurality of scan lines further comprises: a fourth wiring disposed in the display area and that extends in a direction that crosses the second sensor area and is separated from the second wiring by the second sensor area; and a fifth wiring that bypasses the second sensor area and is connected to the second wiring and the fourth wiring, wherein the fifth wiring is spaced apart from the second wiring and the fourth wiring by the first insulating layer.